

WORK EXPERIENCE

Turing | Backend Developer

Dec 2023 - Present

*Worked on B2C Payment, Lending and Notification system features with a Leading Fintech***Payment and Lending System**

- **Designed, build and scaled** major components of lending of loans and collection of repayment via direct and auto payment integrations like AutoPay and Nach. Integrated different events via an event-driven model to ensure robustness and scalability
- **Created a scalable event-based file processing system** to support general org-wide requirements of file processing with a particular use-case of processing file-based payment collection requests for request received from banks

Communication System

- **Designed and developed a centralized notification system** for email, SMS, push and in-app notifications. Added mechanism to improve system's latency, resiliency and reporting. The improvement helped save **INR 50L** a year on SMS
- **Integrated new payment vendors and build org-wide libraries** for dynamic routing to automatically route the notification requests which **reduced vendor cost by 40%**, increased notification delivery guarantees to almost **100%** and reduced customer ticket inflow
- Designed and built a scheduler service for scheduling future events and recurring events. **Built a highly scalable fault tolerant solution** using AWS services like AWS lambda, DynamoDB, SQS, Kinesis streams to achieve a **scale of 1M RPM**

Project Leadership

- Mentored 10+ juniors while working on two different teams. Took ownership of major architectural initiatives within team and created org-wide impact with creation of multiple libraries. Contributed to the hiring process by taking data-driven decisions by refining the technical questions, adding new ones, standardizing the questions and restructuring interview rounds

Sutherland | Backend Developer

May 2022 - Dec 2023

*Worked on B2B Marketing Tools for a US based SaaS company***CRM Marketing Automation Tool**

- Led migration of CRM tool from Heroku to AWS EKS, migrating **15 microservices** and databases leading to **\$5k/month cost reduction**
- Scaled an existing CRM marketing automation tool to handle data loss, ensure consistency, and increase availability of services and auto-scaled infra for variable load for **21k B2B clients with peak scale of 10M transactions per day**
- **Designed and published** an org-wide **robust API migration interface** using sampling of traffic to check disparity between API versions
- **Optimization of 17TB Postgres DB**, tuned vacuum parameters, optimized work mem, indexes, created playbooks for bloat reduction
- **Improved observability of the system** by creating NR dashboards and data warehouse migration of all data and adding visualization

Project Leadership

- Systemized the processes for the CRM tool which was a result of acquisition. Led creation of architectural, sequence diagrams and documentation of major processes. Created on-call playbooks and working runbooks for issues and functionalities respectively. Mentored QAs to work on the backend related tasks

Epam Systems | Backend Developer

May 2019 - May 2022

*Worked on B2C Identity System for a US Fintech firm***Identity System**

- Led integration of TOTP based 2FA, Web-Authn, passkey-based identity verification system for login and transaction flow to perform user identity verification in a **more user-friendly, secure and cost-efficient manner**. The web and app authentication which previously supported just one way of logging in i.e., using a username and password as first factor and using SMS for second factor authentication could now support TOTP and Web-Authn for 2FA, **saving cost**. The two-factor authentication process can now be done in a single step using passkey-based authentication which **adds better user experience** and **prevents them against SIM swapping frauds**
- Used W3C and FIDO standard documentation, RFCs and got internal security review and penetration testing done by the security team. With passkey-based implementation, **reduced the account takeover attempts and slashed costs for SMS based authentication**. **Expenditure on Auth alone got reduced by 10k\$ per month**
- **Synchronized the KYC process** which was previously done in **asynchronous manner making new users wait** for using the application for trading. This solved for the wait between joining to trading on the platform which was previously 1 day
- Supported the org-wide goal of **high fault tolerance and auto-recovery from failures** for the overall ecosystem, lead various initiatives like local integration of systems using Docker, Kubernetes, Wire-mock. Added system safety checks, SLO and smoke tests for API endpoints and UI flow using new metrics, automation scripts, Playwright respectively.

TECHNOLOGIES USED

DSA, Data Structures, Algorithm, System Design, LLD, HLD, Multithreading, Microservices, Distributed Systems, Concurrency, Parallelism, Scalable architecture, Test driver development TDD

Languages: Java, Ruby on Rails, Scala, Golang

Frameworks: Spring Boot, Hibernate, J2EE

Databases: SQL and NoSQL, Postgres, MySQL, Oracle, MongoDB, Elasticsearch, DynamoDB, Cassandra

Cloud: AWS, AWS Lambda, SQS, SNS, S3, Kinesis, EKS, Redis, Heroku Kafka

Open-source frameworks: Kafka, Git, Maven, Docker, Kubernetes, JUnit, Mockito, Flyway, Local-stack, Wire-mock

EDUCATION

DCRUST | B.TECH

May 2015 – May 2019

- Bachelor of Technology, 2019, CGPA: 8